AI500: Foundations of Artificial Intelligence

# Assignment 1
# AI Landscape and Computational Foundations

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## Assignment Overview

This assignment bridges the conceptual and computational foundations of artificial intelligence.

In the theoretical section, you will navigate the end-to-end AI lifecycle by translating ambiguous real-world scenarios into structured AI problem statements, authoring standardized Model Cards and Datasheets for datasets, and conducting systemic failure analyses to identify hidden technical debt, bias, and operational vulnerabilities in production systems.

In the programming section, you will implement foundational mathematical principles from scratch in Python. This includes performing vector and matrix operations, calculating probabilities, and understanding the basics of optimization as well as algorithmic complexity.

## Learning Objectives

Upon successful completion, you will be able to:

- Evaluate real-world AI deployments to identify pipeline architectures, problem classifications, and the root causes of algorithmic and ethical failures.
- Define AI tasks from real-world problems with all the required components.
- Prepare model cards for a given AI system.
- Prepare datasheets for a given dataset.
- Apply mathematical and computational concepts through Python programming to solve AI problems.

# Section 1: Theory (AI Landscape and Problem Framing) — 80 Points

## Section 1 Deliverables

Students are required to submit a single consolidated PDF document containing the answer to the 4 problems detailed below. Detailed instructions and blueprints on how to tackle each questions are provided in the instrctor's notes given on LMS.

## Problem 1: Formulating an AI Problem (20 Points)

The Regional Youth Development Authority (RYDA) oversees employment initiatives for recent high school and college graduates. The authority's leadership proposes the following project:

> **"Build an AI model that predicts which registered candidates are at high risk of remaining unemployed for more than six months without specialized intervention."**

The authority has access to historical data from past cohorts, including candidate demographics, geographical location (urban versus rural), household income bracket, formal education history, and self-reported informal skills (e.g., community volunteering or caregiving). The dataset also includes scores from an initial career aptitude test, the number of interview callbacks candidates receive during their job search, and their verified employment status at the six-month mark.

Predictions could be generated at the time of initial portal registration, immediately following a mandatory one-day orientation, or at the three-month mark of the candidate's job search.

The RYDA operates under strict budget constraints and only has funding to provide 300 candidates per cycle with a fully-funded, high-touch "Digital Skills and Artificial Intelligence Bootcamp." The career placement team must decide which candidates should receive a seat in this intensive bootcamp, access to standard self-paced online modules, a one-time resume-writing workshop, or standard automated job board alerts.

Currently, the authority allocates the 300 intensive bootcamp seats using a first-come, first-served web registration system. This baseline approach often

results in seats being claimed by privileged candidates with reliable internet access, while candidates from marginalized or rural backgrounds miss out. If a candidate who does not actually need the program takes a seat, expensive resources are wasted. Conversely, if a highly vulnerable candidate is overlooked, they may fall into long-term systemic unemployment.

Based on the youth employment scenario provided above, develop a complete AI problem formulation.

**Part A: The Problem-Framing Worksheet**

Complete the following statement by replacing the bracketed text with your specific definitions for this project:

> "At [**decision time**], for [**target population or unit**], using only [**available information**], produce [**required output**] concerning [**prediction horizon or outcome period**], so that [**decision-maker**] can [**action**]. The system will be compared with [**baseline**] and evaluated using [**metrics**], while accounting for [**error costs, subgroup risks, operational constraints, and governance requirements**]."

**Part B: Justification and Analysis**

After completing the statement, briefly explain your reasoning by answering the following:

1. Why is your selected decision time operationally appropriate, and which specific variables in the historical dataset would cause temporal leakage if used at this index time?
2. How precisely should "high risk of remaining unemployed" be defined as a measurable outcome?
3. How does the strict 300-seat capacity limit dictate the type of output your model should produce (e.g., a binary classification, a probability score, or a ranking)?
4. What are the real-world consequences of false positives and false negatives in this specific context, and which demographic or geographic subgroups require separate fairness and performance analysis?

## Problem 2: Preparing a Model Card (20 Points)

A US-based development team has created an artificial intelligence-powered mock interview web application named WePrep. The application uses a multimodal deep learning pipeline — combining a Convolutional Neural Network for facial emotion detection and a text-based model for audio transcript sentiment analysis — to analyze video and audio feeds and provide automated feedback on a candidate's performance. The system was trained on 15,000 recorded mock interviews collected over three years from undergraduate students at a single North American university, and evaluated on an additional 3,000 local interviews alongside a sample of 500 interviews from a European partner university. It processes a downsampled user video feed and real-time speech-to-text transcriptions to output probability scores for six emotional states and an aggregate "Interview Readiness" score ranging from 0 to 100. Overall, the emotion classification achieves 84 percent accuracy with an $F_1$ score of 0.81. However, evaluation logs reveal significant subgroup disparities: accuracy is 89 percent for native English speakers but falls to 62 percent for non-native speakers with accented speech. Furthermore, facial emotion detection accuracy is 87 percent for users with lighter skin tones, 68 percent for users with darker skin tones, and drops to 55 percent for users wearing glasses or sitting in low-light environments.

Prepare a complete Model Card for the WePrep system. Because several operational and ethical details are intentionally omitted from the scenario above, you must critically deduce them based on the provided architecture, data provenance, and performance metrics. Your Model Card must contain:

1. Model name, version, type, and architecture. You must also hypothesize the appropriate purpose, intended role, and intended users for this system.
2. Training and evaluation data, inputs and outputs, overall performance measures, and subgroup performance.
3. Based on the data provided, deduce the likely calibration findings (e.g., when might the model be overconfident?), fairness observations, and known limitations.
4. Define the appropriate uses and specifically hypothesize inappropriate or prohibited uses. Identify potential harms resulting from false positives and false negatives, and outline human-review requirements.

5. Note any generalizability limitations, monitoring requirements after deployment, and conditions that should trigger recalibration, retraining, rollback, or withdrawal.

Conclude the Model Card with a clear deployment recommendation: approve, conditionally approve, or do not approve.

## Problem 3: Preparing a Datasheet (20 Points)

An agricultural research institute has released a large-scale computer vision dataset designed for identifying crop diseases from leaf images. The dataset contains 80,000 images covering four major crops —– wheat, maize, tomato, and potato — across twelve specific disease categories and one healthy baseline category. The images were collected exclusively from 35 large-scale commercial farms, with 70 percent of the total dataset originating from just two geographic regions. The majority of the images were captured during optimal daylight conditions using high-resolution digital cameras, while only 5 percent were taken using low-cost mobile phone sensors. Disease labels were initially assigned by on-site field technicians, with only a 10 percent random sample independently verified by certified plant pathologists. During collection, diseased plants were photographed at a significantly higher frequency than healthy plants. Prior to release, all images were resized to 512 by 512 pixels with their natural field backgrounds left intact, and farm identifiers were replaced with anonymized coded values. The data collection period did not overlap with any seasonal droughts, and no images were sourced from smallholder or subsistence farms.

Prepare a comprehensive Datasheet for this dataset. Your Datasheet must:

1. Hypothesize the motivation for creating the dataset and document its composition, including the unit of observation, geographic and seasonal coverage, and crop and disease distributions.
2. Detail the data-collection process, sampling procedure, label definitions, labeling and verification procedures, and preprocessing transformations.
3. Critically deduce any known missing populations and conditions, potential proxies and dataset artifacts, and possible class imbalances or selection biases.
4. Propose recommended uses for the dataset and explicitly define uses that are

not supported. Note any generalization limitations and explicitly evaluate the risks associated with deploying a model trained on this data to low-cost mobile phones.

5. Hypothesize necessary considerations for consent, privacy, licensing, maintenance, correction, versioning, and retirement procedures.

Conclude by providing a definitive assessment of whether this dataset is suitable for training an AI system intended to be used by smallholder farmers operating in drought-prone regions.

## Problem 4: Identifying Technical Debt and Failures (20 Points)

In 2014, a major multinational technology corporation built an experimental artificial intelligence hiring tool to automate the search for top talent. The objective was to create a "holy grail" engine where recruiters could input 100 resumes, and the system would automatically spit out the top five candidates to hire by scoring them on a scale of one to five stars.

To train the machine learning models, the engineering team used the company's own historical data: successful and unsuccessful resumes submitted over the previous 10-year period. Because the tech industry is historically male-dominated, the vast majority of the resumes in this historical training set came from men. By 2015, the company realized its new system was not rating candidates in a gender-neutral way. The system had taught itself that male candidates were preferred. Specifically, it actively penalized resumes that included the word "women's" (such as "women's chess club captain") and systematically downgraded graduates from two prominent all-women's colleges.

To salvage the project, the engineering team attempted to manually edit the program's code to make those specific terms neutral. However, they soon realized they could not guarantee the tool wouldn't teach itself to find other subtle proxies for gender, such as specific verbs or vocabulary highly correlated with male applicants. Furthermore, because the model was trained on historical human decisions, it was essentially designed to replicate and automate the biases of past recruiters. Executives ultimately lost hope in the project, and the team was disbanded.

Evaluate the real-world case study described above. Conduct an end-to-end failure analysis synthesizing concepts of data provenance, algorithmic bias, and machine learning technical debt. Prepare a failure analysis report documenting the following:

1. Explain exactly how using historical hiring decisions as ground-truth labels created a feedback loop. Why did the model learn that male candidates were "preferable" even if gender was not explicitly labeled?
2. Explain the concept of a proxy variable using the all-women's colleges example. When the engineering team attempted to manually "neutralize" specific words in the code, which specific type of ML technical debt were they creating, and why is this a flawed engineering practice?
3. The original goal was for the system to output a definitive list of the "top five" candidates for recruiters to blindly hire based on a star rating. What are the dangers of this interface design regarding automation bias and a lack of uncertainty reporting?
4. What procedural fairness mechanisms (such as applicant appeal channels or human-in-the-loop overrides) were completely absent from the original "holy grail" vision?

Conclude your report by answering the following: If the company had decided to proceed with a controlled pilot of this tool rather than disbanding the team, what specific monitoring requirements and safeguards would have been absolutely mandatory to prevent real-world harm?

# Section 2: Programming (Computational & Mathematical Foundations) — 20 Points

## Section 2 Deliverables

For this section, you are required to complete the provided Jupyter Notebook template (`PA1_Template.ipynb`) available on the LMS. This notebook contains three programming problems that will test your ability to translate theoretical mathematical concepts into functional Python code.

**Problem Overview:**

- **Problem 1: MLP Forward Pass (8 Points):** You will use `numpy` to perform matrix and vector operations, trace the forward pass of a Multi-Layer Perceptron (MLP) with one hidden layer, and implement the ReLU activation function.
- **Problem 2: Bayes' Theorem (6 Points):** You will calculate prior, evidence, and posterior probabilities to implement the mathematical logic behind a Naive Bayes text classifier.
- **Problem 3: Algorithmic Complexity (6 Points):** You will benchmark the inference latency of sequential list searches $O(N)$ versus hash-based set lookups $O(1)$ across varying vocabulary sizes, plot the results, and evaluate the compute-memory trade-offs.

**Execution & Formatting Instructions:**

- Write your code strictly within the designated `### YOUR CODE HERE ###` blocks. Do not modify any of the provided setup code, dataset arrays, or `assert` statements.
- Ensure your full name and 8-digit student ID are entered in the designated setup cell.
- Before submission, you **must** execute your notebook from scratch to ensure there are no hidden state errors. Restart your environment and run all cells sequentially.
- Verify that all required print statements, calculated variables, and the matplotlib plot are clearly visible in your final output before saving.

# Grading Rubric

| Criterion | Weight | Max Points |
|---|---|---|
| **Section 1: Theory** | **80%** | **80** |
| **P1: AI Problem Formulation** | **20%** | **20** |
| Problem-Framing Statement | - | 8 |
| Decision Time & Leakage | - | 3 |
| Outcome Definition | - | 3 |
| Operational Constraints | - | 3 |
| Error Costs & Fairness | - | 3 |
| **P2: Model Card Preparation** | **20%** | **20** |
| Model Details & Purpose | - | 3 |
| Data & Performance Metrics | - | 4 |
| Inferred Vulnerabilities | - | 5 |
| Appropriate Use & Risks | - | 5 |
| Deployment, Maintenance & Recommendation | - | 3 |
| **P3: Datasheet Preparation** | **20%** | **20** |
| Motivation & Composition | - | 4 |
| Collection & Preprocessing | - | 4 |
| Inferred Vulnerabilities | - | 4 |
| Uses & Limitations | - | 4 |
| Governance & Final Assessment | - | 4 |
| **P4: Technical Debt & Failure Analysis** | **20%** | **20** |
| Historical Bias & Feedback Loops | - | 5 |
| Proxy Variables & ML Technical Debt | - | 5 |
| Automation Bias & Workflow | - | 5 |
| Governance, Fairness & Safeguards | - | 5 |
| **Section 2: Programming** | **20%** | **20** |
| **P1: MLP Forward Pass** | **8%** | **8** |
| ReLU implementation | - | 2 |
| z1 and a1 matrix calculations | - | 4 |
| z2 and final_output calculation | - | 2 |
| **P2: Bayes' Theorem** | **6%** | **6** |
| Prior probabilities | - | 2 |
| Total probability (evidence) | - | 2 |
| Posterior probability | - | 2 |
| **P3: Algorithmic Complexity** | **6%** | **6** |
| Synthetic data generation | - | 1 |
| Loop construction and timing | - | 3 |
| Complexity and memory analysis | - | 2 |
| **Total** | **100%** | **100** |

# Submission Requirements

Please submit the following two files directly to the LMS:

1. A single PDF document containing your complete, formally written answers to all Section 1 questions. Name this file: `StudentID_LastName_A1_Report.pdf`
2. Your completed Jupyter Notebook containing all programming exercises from Section 2. Name this file: `StudentID_LastName_PA1.ipynb`

## Submission Notes:

Due: **September 22, 2026, 11:59 PM PKT**
Total Points: **100** (Section 1: 80 Points and Section 2: 20 Points)

- Submit both files individually via the LMS portal. Do not compress them into a ZIP file.
- Ensure your full name and student ID number are explicitly written at the top of your PDF report and within the designated setup cell of your Jupyter Notebook.

# Academic Integrity

- All code and written responses must be your own work.
- Collaboration on conceptual understanding is encouraged, but implementation must be individual.
- The use of AI-generated text in this assignment is strictly prohibited, and any suspicious submissions will be evaluated using Pangram.

**— End of Assignment —**